

MANASA CHEPURI

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SUMMARY

Data Engineer with over four years of experience building production ETL and ELT pipelines, warehouse architecture, and the BI reporting layer built on top of them, mostly on AWS and Snowflake. Comfortable orchestrating pipelines in Airflow, working with Spark and PySpark for distributed processing, and designing dimensional and star schema models that hold up under real reporting load. Recently expanded into LLM and RAG infrastructure, building LangGraph Text to SQL agents and FAISS and Chroma retrieval for internal AI tooling. Background spans supply chain, healthcare operations, and enterprise data warehousing.

EXPERIENCE

Data Engineer | Calsoft

Jan 2026 – Present

- Built AWS Glue and PySpark ETL/ELT pipelines with metadata instrumentation (ingestion timestamps, row counts, SLA drift), orchestrated through Airflow DAGs with automated failure alerting.
- Built a LangGraph ReAct agent for Text-to-SQL querying and automated log triage, cutting analyst escalation turnaround from hours to minutes.
- Benchmarked OpenAI, BGE, E5, and Sentence Transformer embeddings on Top-K recall, semantic precision, and p99 latency to inform FAISS vs. Chroma vector store selection.
- Built Power BI dashboards and SQL/Python reporting workflows tracking LLM adoption and agent performance, with root-cause analysis on abnormal usage trends.
- Hardened LLM pipeline reliability with structured JSON schema validation, exponential-backoff retry logic, and fallback routing.
- Authored deployment and troubleshooting documentation for internal AI agents, reducing engineering support escalations.

Data Engineer | Community Dreams Foundation

May 2025 – Jan 2026

- Built AWS ETL pipelines (Python/SQL) consolidating 6+ operational systems into a unified warehouse with star-schema models, improving ML feature readiness and cutting report query times.
- Partnered with stakeholders to define KPI metrics and reporting standards, improving reporting accuracy across departments.
- Stood up automated data quality gates and CloudWatch monitoring, catching schema drift and null-rate anomalies before they reached BI dashboards.
- Delivered live Power BI dashboards giving stakeholders self-serve, real-time KPI visibility, replacing manual reporting requests.

Data Engineer | Texas Tech University — Health Sciences Center

Oct 2023 – May 2025

- Automated KPI reporting by integrating Zoom and ServiceNow REST APIs on AWS, eliminating 40% of manual reporting effort and cutting data freshness from D+1 to near real-time.
- Authored Airflow DAGs for agent performance data ingestion, handling schema evolution and multi-month backfills across production pipelines.
- Built Power BI dashboards spanning patient volumes, diagnosis trends, appointment metrics, and call center KPIs (inbound/outbound, abandonment, service levels), cutting ad hoc reporting requests ~30%.
- Worked directly with physicians and clinical teams to gather reporting requirements, delivering SQL queries, stored procedures, and dashboards used in clinical and operational decisions.
- Applied ML classification and LLM-based extraction to call transcripts, surfacing insights that informed staffing and QA training decisions.

Data Engineer | Accenture — Hyderabad, India

Jul 2021 – Aug 2023

- Performance-tuned Oracle and Snowflake data warehouses via schema redesign, materialized views, and indexing, cutting p95 query latency ~35%.
- Automated PL/SQL and Python ETL workflows orchestrated via Airflow, cutting manual data ops 40% across 10+ daily jobs.
- Built ASCP demand-forecasting ML pipelines over multi-SKU procurement data, improving forecast accuracy and reducing overstock costs.
- Monitored production batch jobs and performed data validation and regression testing on the Work Order Management application, catching inconsistencies before they reached downstream reporting.

PROJECTS

[Portfolio: mansachepuri.github.io/portfolio](https://mansachepuri.github.io/portfolio)

- Finance RAG Copilot — AWS Bedrock, pgvector, FastAPI, Streamlit: RAG pipeline for natural-language querying of structured financial data with source-grounded output.
- Olist E-Commerce Pipeline — Airflow, PostgreSQL, Docker: restart-safe Bronze-Silver-Gold medallion pipeline from raw transactions to analytics-ready fact tables.

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, R, Scala, Java

Data Processing: Apache Spark, PySpark, Hadoop, Hive, Kafka, Apache Airflow

Cloud & Infra: AWS (Glue, S3, Redshift, EMR, Kinesis, SageMaker, Lambda, Athena, CloudWatch, QuickSight), GCP, Docker, Kubernetes

Databases: Snowflake, PostgreSQL, Oracle, BigQuery, DynamoDB

Data Analytics & BI: Power BI, Tableau, AWS QuickSight, Star Schema, Dimensional Modeling, Medallion Architecture, KPI Reporting

AI / LLM: LangChain, LangGraph, RAG, FAISS, Chroma, pgvector, Langfuse

EDUCATION & CERTIFICATIONS

M.S. Computer Science, Texas Tech University, Lubbock, TX

Aug 2023 – May 2025

AWS Certified Data Engineer (Udemy) · Oracle Cloud Infrastructure Gen AI Professional · ML Specialization (Coursera) · IBM Data Science · Google Data Analytics · LangChain (Udemy)